MiniShop Manual Test Cases

Application	MiniShop v1 (`mini-shop.html`): login, 3-product shop with cart badge, checkout confirmation, logout
Assessment Part	Part A — Manual & Exploratory Testing
Total Test Cases	23 (Edge: 8 · Positive: 7 · Negative: 5 · Boundary: 1 · State / Edge: 1 · State: 1)
Execution Status	All 23 executed on 27 Sep 2026: **18 Pass** · **2 Fail** · **3 Undefined by spec**
Bugs Found	2 (BUG-A01, BUG-A02)
Specification Gaps	10 (SG-01 to SG-10)
Environment	macOS, Chromium 153, app opened over `file://`. Each case was run from a fresh page load, recording the exact on-screen text, alert text and screen visibility. Both failures were also reproduced in Firefox and WebKit.

> **Status key:** **Pass** = matched the expected result · **Fail** = contradicted a requirement (bug logged) · **Undefined** = the requirements do not say what should happen, so the actual behaviour was recorded and a specification gap raised instead of inventing an expected result. **Priority** is the test case's execution priority (P1 highest). Unless stated otherwise, every case starts from a fresh page load, because all app state (cart, failed-attempt counter, lock) lives in page memory.

Coverage

Login	Test cases
Default login screen	TC-01
Valid credentials	TC-02
Invalid password / invalid username	TC-03, TC-04
Empty inputs	TC-05
Case and whitespace variants	TC-06, TC-07
Three failed attempts / account lockout	TC-08
Attempt after lockout (incl. correct credentials)	TC-09
Non-consecutive failures	TC-10
Failures on other usernames	TC-11
Lockout after page reload	TC-12

Cart	Test cases
Product availability, names and prices	TC-13
Add one product	TC-14, TC-17
Add multiple / different products	TC-14, TC-18
Same product multiple times	TC-15
Cart count (badge) before checkout	TC-14, TC-15, TC-20

Checkout	Test cases
Empty-cart checkout	TC-16
Checkout with one item	TC-17
Checkout with multiple items	TC-18, TC-20
Correct item count and total	TC-15, TC-17, TC-18, TC-20
Cart reset after successful checkout	TC-17, TC-18, TC-19
Confirmation screen	TC-17, TC-18
Back to Shop	TC-19

Logout	Test cases
Logout returns to login screen	TC-21
Username / password cleared	TC-21
Cart after logout and re-login	TC-22
No stale error message after logout	TC-23

Test cases

Login (R1–R4)

ID / Priority / Type	Title / Requirement	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status	Notes
TC-01 P1 Positive	**Login screen is shown by default** R1	Fresh page load (all in-memory state reset)	None required	1. Open `mini-shop.html`	Login screen visible with Username field, masked Password field and Login button. Shop and Confirmation hidden	As expected (`type="password"` masks input)	**Pass**	
TC-02 P1 Positive	**Valid credentials open the shop** R2, R3	Fresh page load (all in-memory state reset)	`standard_user` / `secret123`	1. Username `standard_user`, Password `secret123` 2. Click Login	Login screen hidden, shop screen shown, no error text, badge `0`	As expected	**Pass**	Automated in Part D: `login.spec.js`
TC-03 P1 Negative	**Wrong password is rejected** R4	Fresh page load (all in-memory state reset)	`standard_user` / `wrong`	1. `standard_user` / `wrong` 2. Click Login	"Invalid username or password." Stays on login	As expected	**Pass**	Automated in Part D: `login.spec.js`
TC-04 P2 Negative	**Unknown username is rejected** R4	Fresh page load (all in-memory state reset)	`bad_user` / `secret123`	1. `bad_user` / `secret123` 2. Click Login	"Invalid username or password." (message doesn't reveal which field was wrong)	As expected	**Pass**	Automated in Part D: `login.spec.js`
TC-05 P2 Negative	**Empty username and password** R4, R5	Fresh page load (all in-memory state reset)	Both fields empty; then `standard_user` / `secret123`	1. Leave both fields empty 2. Click Login ×3 3. Enter valid credentials, click Login	Step 2: "Invalid username or password."	Message shown as expected. **But empty clicks count as failed attempts:** after 3 empty clicks the account is locked and valid credentials are refused	**Pass**	Raises SG-04.
TC-06 P3 Edge	**Credentials are case-sensitive** R2	Fresh page load (all in-memory state reset)	`Standard_User` / `secret123`; `standard_user` / `SECRET123`	1. `Standard_User` / `secret123` → Login 2. `standard_user` / `SECRET123` → Login	Both rejected (R2 specifies exact values)	Both rejected with the invalid message	**Pass**	Raises SG-03.
TC-07 P3 Edge	**Leading and trailing whitespace is not trimmed** R2	Fresh page load (all in-memory state reset)	`␣standard_user␣` / `secret123`; `standard_user` / `secret123␣` (␣ = space)	1. `␣standard_user␣` / `secret123` → Login 2. `standard_user` / `secret123␣` → Login	Rejected (literal R2)	Both rejected. A copy-paste trailing space locks out a real user	**Pass**	Raises SG-03.

Lockout (R4, R5)

ID / Priority / Type	Title / Requirement	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status	Notes
TC-08 P1 Boundary	**Lockout boundary: messages on attempts 1–3** R4, R5	Fresh page load (all in-memory state reset)	`standard_user` / `wrong1`, `wrong2`, `wrong3`	1. `standard_user` / `wrong1` → Login 2. `wrong2` → Login 3. `wrong3` → Login	Attempts 1–2: "Invalid username or password." Attempt 3 is itself an invalid attempt, so R4 says it shows the invalid message and the lock message starts on the *further* attempts (R5)	Attempts 1–2 as expected. **Attempt 3 already shows "Account locked. Too many failed attempts."**	**Fail**	Bug BUG-A02 (requirement conflict, SG-01). Automated in Part D: `login.spec.js` (attempts 1–2; the attempt-3 message is not asserted until BUG-A02 is decided)
TC-09 P1 Negative	**Correct credentials are refused while locked** R5	Account locked: TC-08 completed in the same page session	`standard_user` / `secret123`	1. `standard_user` / `secret123` → Login	"Account locked. Too many failed attempts." Shop not shown	As expected	**Pass**	Automated in Part D: `login.spec.js`
TC-10 P1 State / Edge	**Non-consecutive failures do not lock the account** R5	Fresh page load (all in-memory state reset)	`standard_user` / `wrong1`, `wrong2`, `secret123`, `wrong3`, `secret123`	1. `standard_user` / `wrong1` → Login 2. `wrong2` → Login 3. `secret123` → Login (shop opens) 4. Logout 5. `standard_user` / `wrong3` → Login 6. `secret123` → Login	Step 5: "Invalid username or password." (only 1 consecutive failure). Step 6: shop opens	**Step 5: "Account locked…"; step 6: valid login refused.** The counter never resets after a successful login	**Fail**	Bug BUG-A01. Automated in Part D: `login.spec.js` (expected-to-fail test documenting BUG-A01)
TC-11 P2 Edge	**Failures on *other* usernames and the valid account** R5	Fresh page load (all in-memory state reset)	`alice` / `x`, `bob` / `x`, `carol` / `x`; then `standard_user` / `secret123`	1. `alice` / `x`, `bob` / `x`, `carol` / `x` → Login each 2. `standard_user` / `secret123` → Login	Undefined: R5 says "the account" locks but doesn't say whose account	`standard_user` is locked. The lock is global to the browser tab, not per account	**Undefined**	Spec gap SG-02.
TC-12 P2 Edge	**Lockout survives a page reload** R5	Account locked (3 failed attempts in the current page session)	`standard_user` / `secret123`	1. Reload the page 2. `standard_user` / `secret123` → Login	Undefined: no lock duration or persistence is specified	**Reload clears the lock** and the login succeeds. The lock is trivially bypassed (see OBS-02)	**Undefined**	Spec gap SG-02.

Product list and cart (R6, R7)

ID / Priority / Type	Title / Requirement	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status	Notes
TC-13 P1 Positive	**Shop lists 3 products** R6	Fresh page load (all in-memory state reset)	Valid login	1. Log in	3 rows: Wireless Mouse $19.99, Mechanical Keyboard $49.99, USB-C Hub $24.99, each with an "Add to Cart" button	As expected	**Pass**	Automated in Part D: `checkout.spec.js`
TC-14 P1 Positive	**Badge starts at 0 and +1 per click** R7	Fresh page load (all in-memory state reset)	Wireless Mouse, Mechanical Keyboard, USB-C Hub	1. Log in 2. Add Mouse, then Keyboard, then Hub	Badge `0` → `1` → `2` → `3`	As expected	**Pass**	Automated in Part D: `checkout.spec.js`
TC-15 P2 Edge	**Same item added multiple times** R7, R9	Fresh page load (all in-memory state reset)	Wireless Mouse ×3	1. Log in 2. Add Wireless Mouse ×3 3. Checkout	Badge `3`. Summary "3 item(s), total $59.97"	As expected (each click adds a separate line)	**Pass**	Raises SG-06. Automated in Part D: `checkout.spec.js`

Checkout and confirmation (R8, R9, R11)

ID / Priority / Type	Title / Requirement	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status	Notes
TC-16 P1 Negative	**Checkout with an empty cart** R8	Fresh page load (all in-memory state reset)	Empty cart	1. Log in (badge `0`) 2. Click Checkout	Browser alert "Your cart is empty." Stays on shop, no confirmation	As expected	**Pass**	Automated in Part D: `checkout.spec.js`
TC-17 P1 Positive	**Checkout with a single item** R9	Fresh page load (all in-memory state reset)	Wireless Mouse ×1	1. Log in, add Wireless Mouse 2. Checkout	Confirmation "1 item(s), total $19.99". Badge reset to `0`	As expected	**Pass**	Automated in Part D: `checkout.spec.js`
TC-18 P1 Positive	**Checkout with one of each product** R9	Fresh page load (all in-memory state reset)	Wireless Mouse, Mechanical Keyboard, USB-C Hub (1 each)	1. Log in, add Mouse, Keyboard, Hub 2. Checkout	"3 item(s), total $94.97" (19.99 + 49.99 + 24.99). Badge `0`	As expected	**Pass**	Automated in Part D: `checkout.spec.js`
TC-19 P1 State	**Back to Shop, and the cart really is emptied** R8, R9, R11	Fresh page load (all in-memory state reset)	Cart from TC-18 (one of each product)	1. Complete TC-18 2. Click Back to Shop 3. Click Checkout	Step 2: shop shown, badge `0`. Step 3: empty-cart alert (the cart array is emptied, not just the badge)	As expected	**Pass**	Automated in Part D: `checkout.spec.js`
TC-20 P3 Edge	**High-volume cart total accuracy** R7, R9	Fresh page load (all in-memory state reset)	Wireless Mouse ×12, USB-C Hub ×7, Mechanical Keyboard ×5	1. Log in 2. Add Mouse ×12, Hub ×7, Keyboard ×5 3. Checkout	Badge `24`. "24 item(s), total $664.76" (hand-calculated in cents)	As expected, with no floating-point drift. A wider sweep (0–30 of each product, 29,791 combinations) also matched to the cent	**Pass**	See OBS-06.

Logout and session (R10)

ID / Priority / Type	Title / Requirement	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status	Notes
TC-21 P1 Positive	**Logout returns to login and clears fields** R10	Fresh page load (all in-memory state reset)	Valid login	1. Log in 2. Click Logout	Login screen shown, shop hidden, Username and Password both empty	As expected	**Pass**	Automated in Part D: `navigation.spec.js`
TC-22 P2 Edge	**Cart after logout and re-login** R10	Fresh page load (all in-memory state reset)	USB-C Hub ×2	1. Log in, add USB-C Hub ×2 2. Logout 3. Log in again 4. Checkout	Undefined: R10 doesn't say whether logout ends the cart	**Cart survives logout.** Badge shows `2`, and checkout confirms "2 item(s), total $49.98"	**Undefined**	Spec gap SG-05.
TC-23 P3 Edge	**No stale error message after logout** R4, R10	Fresh page load (all in-memory state reset)	One invalid login, then `standard_user` / `secret123`	1. Fail one login 2. Log in successfully 3. Logout	Login screen shows no leftover error text	As expected (a successful login clears the message)	**Pass**	

Referenced IDs

Bug	Summary	Severity / Priority
BUG-A01	Lockout triggers on non-consecutive failures: the failed-attempt counter never resets after a successful login	High P2
BUG-A02	The 3rd failed attempt shows the lock message instead of "Invalid username or password." (R4/R5 conflict, needs a PO decision)	Low P3

Observation	Summary
OBS-02	Lockout is enforced only in page memory, so a reload or DevTools bypasses it
OBS-06	Money is summed as floating point (correct today, but fragile)

Full bug reports and all 6 observations: `bug-reports.xlsx`. Full specification-gap details: `test-plan.pdf`, section 7.

Spec gap	Summary
SG-01	R4 vs R5 conflict: which message the 3rd failed attempt shows
SG-02	Lock scope, duration and unlock path are undefined
SG-03	Credential matching rules (case sensitivity, whitespace trimming, limits)
SG-04	Empty-field handling, and whether empty submits count toward the lock
SG-05	Whether logout ends the session's cart
SG-06	Cart model (units vs products, maximum quantity, view/remove items)
SG-07	Confirmation content is under-specified
SG-08	No out-of-stock state in the UI requirements
SG-09	Session persistence (reload, back, new tab, Logout on confirmation)
SG-10	No non-functional requirements (accessibility, browsers, security)